

KRISHNA Koushik Thokala

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Summary

Infrastructure operations engineer with 2 years leading five-engineer shifts in a 24/7 carrier NOC, coordinating P1/P2 response for 25–30 Tier-1 clients against a 99.9% availability target and four-hour restoration commitments. Built AWS platforms with Terraform, EKS, GitOps, serverless services, and observability tooling, including a live monitoring system and validated failure-recovery workflows.

Certifications

AWS Certified Solutions Architect – Associate | Amazon Web Services | July 2026

Technical Skills

Cloud & IaC: AWS, Terraform, VPC, EC2, EKS, ECR, Lambda, RDS, DynamoDB, S3, API Gateway, CloudFront
Containers & Delivery: Kubernetes, Helm, Docker, Argo CD, GitOps, GitHub Actions, CI/CD, Checkov, kubeconform
Security & Access: IAM, OIDC, RBAC, KMS, Secrets Manager, Security Groups, NetworkPolicies
Observability & Reliability: Prometheus, Grafana, CloudWatch, X-Ray, k6, HPA, SLIs/SLOs, burn-rate alerting
Systems & Networking: Python, Bash, Linux, Git, AWS CLI, TCP/IP, HTTP/S, DNS, BGP, MPLS

Projects

EKS Reliability Platform | EKS, Terraform, Helm, Python/FastAPI, RDS **GitHub**

- Provisioned a multi-tier AWS environment with Terraform, including public, private, and database subnets, managed EKS, RDS PostgreSQL, ECR, Secrets Manager, IRSA, and CloudWatch monitoring.
- Implemented six Prometheus availability and latency recording rules with multi-window burn-rate alerts against a 99.9% SLO, validated through five promtool tests.
- Load-tested the EKS-hosted backend with 18,819 requests and zero failures or pod restarts; HPA scaled the service from 2 to 6 replicas under peak load and returned it to baseline afterward.

GitOps Delivery Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions **GitHub**

- Built an EKS GitOps platform from 15 reusable Terraform modules, with four Argo CD Applications reconciling dev, staging, prod, and observability namespaces under namespace-scoped RBAC, quotas, and network policies.
- Secured image promotion with a four-gate GitHub Actions workflow that validates Git and ECR provenance, authenticates through branch-scoped OIDC without stored AWS credentials, and delivers each promotion through a pull request.

Serverless Uptime Monitor | Lambda, Python, DynamoDB, Terraform **Live | GitHub**

- Deployed a live serverless uptime platform with scheduled Lambda checks and state-change alerting; hardened it with API throttling, dead-letter queues, point-in-time recovery, KMS encryption, CloudWatch monitoring, and X-Ray tracing.
- Designed a latest-status access pattern that reads 10 current records instead of scanning 86,400 retained records in the reference workload, keeping dashboard-read cost constant as history grows.
- Validated stateful alerting through an outage drill with one-second failure detection, dashboard visibility within 26 seconds, duplicate-page suppression, and recovery notification.

Experience

Tata Communications **July 2022 – July 2024**
Senior Engineer, Network Operations Center – Shift Lead *Pune, India*

- Led five-engineer shifts in a 24/7 carrier NOC, coordinating triage of 40+ daily incidents across 25–30 Tier-1 clients and prioritizing concurrent P1/P2 outages against 99.9% availability and four-hour restoration commitments.
- Diagnosed BGP and MPLS routing failures, TCP/IP connectivity faults, DWDM impairments, and hardware failures across Juniper, Huawei, Ciena, and Alcatel infrastructure.
- Directed incident restoration through ServiceNow, coordinating vendor TAC escalation, field dispatch, customer-premises testing, and client communication throughout active outages.
- Produced post-incident RCAs and cross-shift handoffs that preserved diagnostic context and supported recurrence prevention; earned a Certificate of Excellence for resolving an escalated customer incident.

Education

Indiana University Bloomington **August 2024 – May 2026**
Master of Science in Computer Science, GPA: 3.5/4.0 *Bloomington, IN*

SRM Institute of Science and Technology **June 2018 – June 2022**
Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10 *Chennai, India*